

Sahil Upadhyay

SENIOR ENGINEERING LEAD | FULL-STACK & AI

Java · Spring Boot · React/TypeScript · Kafka · AWS
Now: leading an HPE platform processing 1B+ files a day

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PROFILE

Senior Engineering Lead and hands-on full-stack engineer (Java/Spring Boot, React/TypeScript, C/C++, Kafka, AWS) with **8+ years** across enterprise tech, automotive, banking, media, and logistics. Currently leading engineering on an **HPE platform that processes 1B+ files a day**; previously delivered **85% faster page loads**, **90% fewer deployment errors**, and **99.9% database availability**. Ships AI to production – assistants, RAG, MCP servers, agentic workflows – and trains US and European engineering teams to do the same.

01 EXPERIENCE

• TEKsystems

Feb 2025 – Present

Senior Engineering Lead – HPE and Mercedes-Benz engagements

📍 Bengaluru, India

🔧 C/C++ · Linux · Java · Spring Boot · RESTful APIs · React/TypeScript · Kafka · Liquibase · SQL · Splunk · CI/CD · MCP · Agentic AI

HPE · RDA platform – 1B+ files processed per day

- **Lead engineering for a platform that ingests and processes 1B+ files a day** – C/C++ services on Linux in the RDA core, Java/Spring Boot services, and React consoles – sustaining throughput through memory pooling, batched I/O, bounded concurrency, and back-pressure.
- **Trained HPE engineering teams across the US and Europe on integrating AI into existing workflows** – prompting techniques, multi-agent orchestration and parallel development with Cursor and Claude, building MCP servers, and agentic workflows through VS Code plugin development.

Mercedes-Benz · Automotive – dealer contract legality and compliance

- **Led Java/Spring Boot and React delivery for the dealer-contract compliance platform** – contract mass-upload, signatory verification, duplicate prevention – from ambiguous legal requirements to auditable production features signed off with product, legal, QA, and operations.
- **Eliminated repeat production incidents and cut API/query latency** by turning root causes into permanent validation rules and release gates, adding composite indexes, keyset pagination, and batch writes, and shrinking React bundles with code splitting and lazy loading.

• Persistent Systems Limited

Aug 2024 – Feb 2025

Engineering Lead – Banking client, digital lending and servicing

📍 India

🔧 Java · Spring Boot · Node.js · React · Hibernate · Kafka · Redis · SQL/NoSQL · AWS · Jenkins · Docker · Kubernetes

- **Led React and Spring Boot delivery for a retail-banking platform**, streamlining loan origination, KYC, and servicing journeys so customers complete applications in fewer steps and staff close cases faster.
- **85% faster page loads** across loan and account screens – query-plan tuning, indexing, N+1 elimination, and Redis caching paired with memoization, virtualized rendering, and code splitting.

- **80% faster releases, 90% fewer deployment errors, 75% less server downtime** for the banking app by standardizing Jenkins CI/CD pipelines, Docker images, and Kubernetes rollouts with health checks and rollback-safe releases.
- **Kept real-time transaction and document ingestion available at peak load** with partitioned Kafka flows and idempotent consumers on AWS, and cut incident-investigation time with an AI-assisted analysis workflow over Splunk and JProfiler telemetry – every fix human-validated.

Capgemini

Oct 2021 – May 2023

Systems Architect (Professional 2) – The New York Times, global newspaper delivery

India

Java • Spring Boot • RESTful APIs • React • Kafka • Docker • Kubernetes • Cloud migration • CI/CD • Splunk

- **Built and maintained subscriber-delivery features for The New York Times' global newspaper delivery application** – Spring Boot REST APIs and React – expanding what the business could sell and service across international delivery markets.
- **80% more reliable releases and 80% fewer cross-OS discrepancies** by defining reusable architecture, coding, automated-validation, containerization, and release standards adopted across delivery teams.
- **Standardized service boundaries, API contracts, retry policies, and health checks** across REST APIs, React surfaces, Kafka integrations, and Docker/Kubernetes cloud migrations with CI/CD.
- **Removed database and JVM throughput bottlenecks** on delivery and subscription workloads, diagnosed via execution plans, index selectivity, connection-pool metrics, and Splunk traces, fixed with batching, caching, and async events.

GroupFIO

May 2021 – Aug 2021

Fullstack Developer – Customer-support product, built from scratch

India

Java • React • Node.js • Express • SQL • REST APIs

- **Designed and built a customer-support product from a blank page** – ticketing, conversation history, and agent workspace – across the React front end, Node.js/Express API layer, and Java services.
- **50% faster API response times** by tuning the Java path end to end: connection pooling, prepared-statement and query optimization, response caching, asynchronous processing, and N+1 removal.
- **Shipped a responsive, low-friction agent interface** with designers and stakeholders, cutting the clicks needed to resolve a ticket and keeping support flows smooth under concurrent load.

Kent RO Systems

Nov 2020 – Apr 2021

Senior Analyst (Contract)

India

Neo4j • Java • Kafka • AWS • Docker • MySQL • MongoDB • RabbitMQ • Jenkins • PowerShell

- **Led development of a data catalog and lineage system** on Neo4j, Java, and Docker, integrating Kafka and AWS for scalable infrastructure and stronger data integrity.
- **85% better page responsiveness** in front-end modules with real-time analytics, tuned with VM Profiler.
- **80% less manual administration** by automating operations with PowerShell/Shell scripting and Jenkins CI/CD, while running AWS data pipelines across MySQL, MongoDB, RabbitMQ, and Kafka.

Zyla Clothing

Mar 2020 – Nov 2020

Founder & CEO – Direct-to-consumer clothing startup

India

Java • HTML/React • Shopify plugins • Generative AI • Google Ads • SEO

- **Founded and built the Zyla storefront end to end** – Java services with an HTML/React front end and Shopify plugins for catalog, checkout, and order flows – turning a clothing idea into a live online store.
- **75% higher sales conversion** through UI/UX optimization and personalization, with a storefront fast enough to hold the traffic that paid campaigns brought in.
- **Generated 400K+ impressions** by running Google Ads and SEO on top of an SEO-ready build – structured markup, fast page loads, clean URLs – so marketing spend landed on pages that ranked and converted.
- **Automated customer support and product recommendations** with generative-AI tools, lifting engagement and satisfaction – my first AI feature in front of paying customers.

● NIIT Technologies

Jan 2020 – Mar 2020

Software Developer – Library management system for US colleges

📍 India

🔗 Java • Spring Framework • Kafka • Agile

- **Built Java/Spring features for a library management system used by colleges across the US** – catalog, circulation, and member workflows – delivering **30% faster** through cross-functional Agile teamwork.
- **Integrated Kafka for real-time catalog and circulation updates**, improving responsiveness and reliability.
- **Raised code quality across the team** through code reviews and mentoring of junior developers.

● Cognizant

Dec 2018 – Dec 2019

Programmer Analyst – UPS, package delivery and tracking

📍 India

🔗 Java • Spring Boot • Node.js • HTML5/JavaScript • RESTful APIs • MySQL • Kafka • Hadoop • Hive • Spark • Jenkins • Splunk

- **95% faster data processing** on UPS shipment and tracking data pipelines built with Hadoop, Hive, and Spark, integrating Kafka for real-time package-event processing.
- **85% higher throughput and 60% lower response latency** on the high-traffic delivery application by optimizing the HTML5/JavaScript front end and Java/Spring Boot back end with design patterns and SOLID principles, and building RESTful APIs (Java, Node.js, MySQL) deployed through Jenkins.
- **Near-zero discrepancies in package and delivery data** through algorithmic big-data validations and Splunk monitoring, keeping tracking information trustworthy at scale.

● Wipro

May 2017 – May 2018

Associate Analyst

📍 India

🔗 SQL Server • Database design • Indexing • Backup & recovery

- **99.9% uptime** for enterprise SQL databases through robust backup and recovery strategies that cut data-loss risk and ensured business continuity.
- **70% faster queries** through database design, indexing, and query tuning on SQL Server.
- **Administered enterprise-grade SQL databases** for high availability, performance, and security.

02 SELECTED AI AND PLATFORM PROJECTS

MultiAgentOPENAI Python • OpenAI

Multi-agent orchestration framework – specialized agents, tools, handoffs, and guardrails.

Replico AI Python • Vision

Natural-language to desktop-automation agent with vision targeting, replay, and emergency stop.

3DCodePad React • Three.js • Monaco

Browser IDE running C, C++, Java, Python, and JavaScript through a remote execution API.

Multi-Agent Iterative AI System ChromaDB • RAG

Offline LLM refinement loop with RAG over a ChromaDB vector store; containerized.

NearNest Next.js • Firebase • OpenAI

Local-services marketplace – passwordless auth, geospatial discovery, payments, OpenAI assistant.

Data Lineage Catalog Neo4j • Java • Kafka

Neo4j lineage graph across SQL/NoSQL and cloud services, fed by Kafka events.

03 EDUCATION AND CERTIFICATIONS

Lovely Professional University

2014 – 2018

B.Tech, Electronics and Communication Engineering • Minor: Android Development

📍 India

Security

Java Advanced Application Security (2025) • OWASP Top 10 • Container Hardening (Docker)

Data

Advanced Data Analytics (2023) • Business Intelligence (2023) • Big Data – Hadoop 2.x (2019)